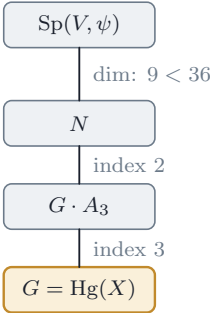


$$M = \mathbb{G}_m \cdot M^1, \\ M^1 = M \cap \mathrm{Sp}(V, \psi)$$

dim of the algebraic classes in $V^{\otimes 2m}$

X^4 $m = 2$	X^6 $m = 3$	formula, $c = 2$ or 5	$\zeta \in M^1$	Det, r_1 algebraic	exceptional classes algebraic
3	15	$(2m - 1)!!$	yes	no	no
4	35	$(c^3 + 3c^2 + 2c)/6$	yes	yes	no
4	45	$(c^3 + 2c)/3$	yes	yes	no
8	125	c^3	no	yes	yes



$M^1 = G \Leftrightarrow$ the Hodge conjecture for every $X^n \Leftrightarrow$ the two exceptional classes of $X \times X$ are algebraic
 $\Leftrightarrow$ some algebraic class on some power of X is not fixed by ζ

subgroups of $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}}) \cong S_3$;
dashed: not stable under the Galois group

